

Marco Fontana

Junior Electronic Engineer

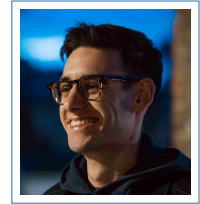
Castel d'Ario – Mantova

* 24 january 1997

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github.com/its-fonsy



Education

- 2020–2024 **Master's Degree in Electronic Engineering**, 100
Alma Mater Studiorum - University of Bologna
Master Thesis *Hardware and Software development of an Ethernet module on FPGA*
- 2016–2020 **Bachelor's Degree in Computer Science, Electronic and Telecommunication Engineering**, 104
University of Parma
Thesis *Performance analysis of an IP module for FFT computation on FPGA*
- 2011–2016 **Secondary School certificate in Technical Institute of Technology**, 94
Istituto Superiore Enrico Fermi – Mantova
Thesis *Rubik's cube solver robot*

Master Thesis

I have contributed to AlSaqr project, an Open Source SOC that embeds a RISC-V 64-bit CPU and other peripheral to be used on drones. My job was to implement an Ethernet IP inside the SOC and verify it with FPGA board (VCU118 and Genesys2) and Linux OS (CVA6-SDK).

References

AlSaqr: <https://github.com/AlSaqr-platform/he-soc>

CVA6-SDK: <https://github.com/AlSaqr-platform/cva6-sdk>

Experiences

- from 05/2024 **Electronic Engineer**, Fraste, Nogara (VR)
to 11/2024 Development of electrical system for drilling rigs.
- from 04/2024 **NuttX Driver Developer**, University of Bologna, Bologna (BO)
to 05/2024 Development and test of an Ethernet driver for NuttX.
- from 05/2015 **Electrician**, Gasparini P.I. Andrea, Castel d'Ario (MN)
to 07/2015 Electrical system for houses

Computer skills

Programming Languages

VHDL, SystemVerilog, C, C++, Bash,
Python, AVR-Assembly

Operating Systems

GNU/Linux (Arch, Ubuntu, Debian, Fe-
dora), Windows, Mac OS

Office Applications

Word, Excel, PowerPoint

Development Systems

Git, Make

3D Modeling Softwares

FreeCAD, OpenSCAD, Fusion360

Markup Languages

CSS, HTML, $\text{\LaTeX}$, Markdown

Languages

Italian Native

English B2 (read, written and spoken)

Interests

- Books: science fiction, fantasy, classics, etc.
- 3D Printing and Do It Yourself projects;
- Technology;
- Videogames;
- Computers;
- Open Source and self-hosting;

Personal Projects

Library for light sensor TSL2591

<https://github.com/its-fonsy/tsl2591-avr>

Library written in C for the light sensor TSL2591 aimed to be used with ATmega328P.

Pomodoro timer (AVR)

<https://github.com/its-fonsy/avr-pomodoro>

Pomodoro timer realized with Atmega328p and OLED display SSD1306. The library to control the OLED-display and the microcontroller peripheral has been implemented from scratch.

Pomodoro timer (STM32)

<https://github.com/its-fonsy/stm32-pomodoro>

Implementation of "Pomodoro timer (AVR)" project using STM32F401RCT6 (Black Pill) microcontroller. The development has been made inside STM32CubeIDE.

Lyrpy

<https://github.com/its-fonsy/lyrpy>

Python application that prints on the terminal the lyrics of the playing song and it emphasize the verse that is going singed.

Fastfingers

<https://github.com/its-fonsy/fastfingers>

A replica for the terminal of the "10FastFingers" games written in C using ncurses library.